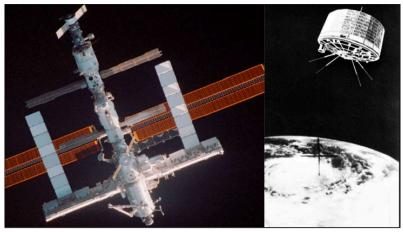
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After completing this session, you will be able to explain the interdependency among

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Technology, environment, society and development

- ✓ **(the study and knowledge of) the practical, especially industrial, use of scientific discoveries:**
  i.e. computer/medical/space technology.
- ✓ **the methods for using scientific discoveries for practical purposes,**
- ✓ **esp. in industry**
- ✓ **the use of scientific knowledge or processes in business,**
- ✓ **industry, manufacturing, etc.:**

https://dictionary.cambridge.org/dictionary/english/technology
https://en.wikipedia.org/wiki/Technology_and_society

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Technology, Environment & Society

What is technology?

The term ***technology***, a combination of the Greek ***technē***, "art, craft," with ***logos***, "word, speech," meant in Greece a discourse on the arts, both fine and applied.

Technology is
the application of scientific knowledge
to the practical aims of human life or,
as it is sometimes phrased,
to the change and manipulation of the human environment.

Is Internet technology "making us stupid"?

What is the impact of artificial intelligence (AI) technology on society?

https://www.britannica.com/technology/history-of-technology

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Technology, **Environment**, society and development

Definitions from Oxford Languages

- ✓ **the surroundings or conditions** in which a person, animal, or plant lives or operates.

Ex. "survival in an often hostile environment"

- ✓ **the natural world**, as a whole or in a particular geographical area, especially as affected by **human activity**.

Ex. "the impact of pesticides on the environment"

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Technology, **Environment**, society and development

Environment (NATURE)

- ✓ **the air, water, and land in or on which people, animals, and plants live:**

i.e. We're not doing enough to protect the environment from pollution.

Environment (SURROUNDINGS)

- ✓ **the conditions that you live or work in and the way that they influence how you feel or how effectively you can work:**

Studies show that if a working environment is pleasant, productivity increases.

https://dictionary.cambridge.org/dictionary/english/environment
https://en.wikipedia.org/wiki/Technology_and_society

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Technology, Environment, **Society** and development

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Relationship among Environment, Society and Technology

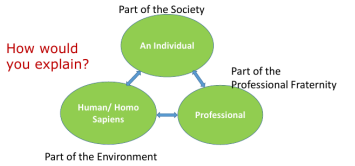
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Relationship among Technology, Environment, and Society

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Relationship among Technology, Environment, and Society

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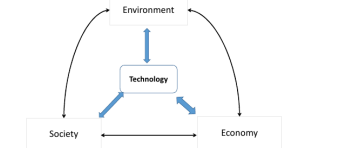


Fig.: Dimension of Sustainable Triangle

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NASA Image, 2012

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Relationship among Environment, Society and Technology

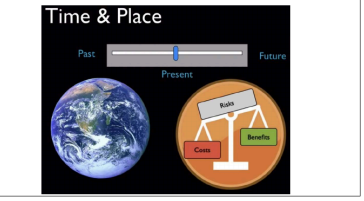
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Environment and **Earth System**

Centrosphere

Earth was once a **molten mass** and to this date, the temperature and pressure in the interior of the earth is so high that the molten state still exists there. These are brought to the surface from time to time as **lava**, through **volcanic eruption** or through faults in the earth's crust.

Lithosphere

The lithosphere consists of the **solid crust** on which rest the continental masses comprising of **mountains, valleys and plains**. This is where human beings and other terrestrial animals live. The soil covering the solid masses in most cases is fertile enough to support plant life and in its natural state is covered with vegetation.

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Environment and **Earth System**

Hydrosphere

The hydrosphere consists of these oceans, seas and the inland water bodies. **Ninety-seven percent** of the total supply of water on earth are held in the oceans and seas whilst the rest is freshwater. Life on earth would not be possible without water. All the life supporting processes of the earth system are heavily dependent on water.

Atmosphere

The atmosphere is a gaseous mantle enveloping the earth. The main constituents are nitrogen (78%) and Oxygen (21 %). The rest 1 percent is mostly argon, carbon dioxide and some other trace gases.
It consists of different strata called: Troposphere, Stratosphere, Mesosphere and Thermosphere.

Biota: all living organism including Flora and Fauna

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Ecology and **Ecosystem**

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Ecology and Ecosystem

is the study of **living organisms** in the **natural environment**.
How they interact with **one another** and
how they interact with their **non living environment**.

is the relationship between organism and environment. (Hackel and Tansley)

Terminologies
Autecology: Individual (Study of RB Tiger Population)
Synecology: Group/community (Study of Forest Ecosystem)
Ecological Succession: **process of change** in the species structure over time
Ecological Niche: Physical + Functional **role of a species** in a community
Ecological Equivalent: Similar niche in different geographical regions

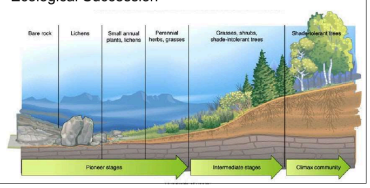
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Ecological Succession



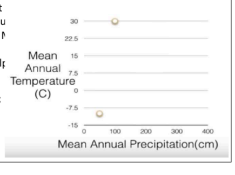
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Ecological Succession

Therms:
- Mega: High, T. Rain Forest
- Meso: High+Low, T. Decidu
- Micro: Low Temperature, M
  Coniferous
- Hekisto: Extremely low, Al

Biomes:
very large ecological areas; climate, vegetation, soil



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The characteristics of the type of environment where an organism normally lives. (e.g. a stoney stream, a deciduous temperate woodland, Bavarian beer mats)

Habitat



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Ecology and **Ecosystem**

A unit which consists of all living organisms functioning together in a given area and interacting with the physical environment in a manner, whereby, flow of energy leads to clearly defined biotic structures and cycling of materials between living and non-living parts, is know as an ecosystem.

- Ecosystem, the term coined by Tansley
- Smallest structural and functional unit of nature
- **Consisting of both biotic community and the abiotic environment which have close interactions**
- The interaction is conducted by **energy flow** (**Solar energy**) in the system and cycling of materials (**Nutrient cycle**)
- Entire biosphere is considered as a global ecosystem

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Ecology and **Ecosystem**

https://www.britannica.com/science/ecosystem

An ecosystem can be categorized into its abiotic constituents, incl. minerals, climate, soil, water, sunlight, and all other **nonliving elements**, and its **biotic constituents**, consisting of all its living members.

Two major forces link these constituents together:
(i) the flow of energy through the ecosystem and
(ii) the cycling of nutrients within the ecosystem.



Community + Abiotic environment, interacting

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Ecology and **Ecosystem**

Characteristics of Ecosystem

Boundaries are indistinct and over-lapping; could be temporary or permanent; even may be small (drop of water) or large (whole earth)

Types of Ecosystem: on the basis of the origin

1.**Natural** Ecosystem; terrestrial (forest, grassland and desert) and aquatic (pond, lake, wetland, river and estuary).

2. **Man-made** Ecosystem; crop field and aquarium.

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Ecology and **Ecosystem**

Types of Ecosystem: **on the basis of the size**

- ❑Mega Ecosystem: Entire Biosphere
- ❑Macro Ecosystem: Major Biomes like Tropical Rain forest (Amazon)
- ❑Micro Ecosystem: Pond
- ❑Nano Ecosystem: Drop of water

COMPONENTS OF ECOSYSTEM

1. **Biotic Components** includes living component from smallest microbe, fungi to largest animals and plants
2. **Abiotic Component** includes Inorganic substance (carbon, nitrogen, CO_2, H_2O etc.) involved in natural cycles, air, water and other physical factors, organic compounds (Proteins, Carbohydrates, Humus)

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Ecology and **Ecosystem**

Among all abiotic components; the important ones are:

- **Temperature**; spatial distribution of species decreases on either side of equator towards the poles and as we go above and below the MSL.
- Water; Life started in water about 3.5 billion years ago.
- Light; photosynthesis
- Soil; Edaphic factors. Based on formation process, it is classified in two types:

  Residual soil – Weathering and pedogenesis occur at the same place
  Transported soil - Weathering and pedogenesis occur at different places (Colluvial- Brought by gravity, Alluvial- brought by water, Glacial- by snow, Eolian- by wind)

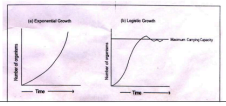
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Carrying Capacity

Carrying capacity of environment or an ecosystem is the **threshold limit** of use of that system without damaging it.
Every ecosystem has its resources that are used for economic development, for survival and for habitat creation. Environment and ecosystems have got the abilities to recover the loss of its resources by regenerating them over the period of time that are temporary and not exceeding the threshold damage limit.



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Carrying Capacity

Catton (1986), defined an environment's carrying capacity as its maximum persistently supportable load.

Ecological carrying capacity is a measure of the amount of renewable resources in the environment in units of the number of organisms these resources can support. Normally, K is a function of both the species and the environment, and is expected to change only in evolutionary time (Daily and Ehrlich, 1992).



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Carrying Capacity



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Sustainable development (see other slides);

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Thank You!
Welcoming Q & A

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